Disinformation Detection
State of the Art and Phase 1 Project Proposal

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Abstract

We review fifteen studies and plan to compare lexical classifiers, fine-tuned encoders, and prompted large language models. We will test whether persuasion and intent help models work across Polish and English datasets. All numbers come from published studies; we have not run our own experiments yet.

1 Problem and review scope

The Slavic NLP 2027 task[1] is a binary task: a model decides whether a text contains disinformation. It provides Polish and English data and will also include test data in other Slavic languages.

This review focuses on studies from 2024–2026 and one important study from 2023. We searched the ACL Anthology for work on large language models (LLMs), intent, persuasion, and transfer between languages and domains. We then checked the papers and code repositories. Table 1 summarises the studies, while Table 2 shows their results and evaluation settings.

2 State of the art and research gap

Supervision versus prompting. Polish RoBERTa-large reaches a weighted F1 score of 0.96 on MIPD and an F1 score of 0.93 for the positive class. The study uses a random split, so these results show performance within one dataset (Modzelewski et al., 2024). F3 reports useful zero-shot results on data from a different distribution (Lucas et al., 2023). Because these studies use different data and evaluation settings, we cannot directly rank all model types.

Persuasion and intent. PCoT first looks for persuasion strategies and then predicts disinformation (Modzelewski et al., 2025). MALINT follows a similar idea. It uses intent-based inoculation and tests models in several languages (Modzelewski et al., 2026). These extra steps may help, but persuasive language alone does not mean that a text is false. EU DisinfoTest shows that rhetorical framing can affect model predictions (Sosnowski et al., 2024). The MultiDis guidelines also place some misinformation and disinformation examples in the same class. Therefore, datasets with similar binary labels may still define the task differently.

Robustness and validity. FADED tries to reduce bias related to documents and named entities when the domain changes (Wei et al., 2025). Tests across datasets reveal problems that may be hidden by high accuracy on one dataset (Calvo and Laureano, 2026). TripleFact and SSA show that LLM results may be too high when the model has seen similar benchmark data before (Xu and Yan, 2025; Xu et al., 2025). A recent publication date does not guarantee that a modern model has not seen the test data. The source of the text also matters. Transfer between human-written and machine-written news can work differently in each direction (Su et al., 2024). Fake news written with help from an LLM is also hard to detect (Wang et al., 2025).

3 Data and readiness

Table 3 lists the datasets that we will use. Additionally, if time and resources permit, we may identify and incorporate a dataset containing information in another Slavic language. Before training, we will record the exact dataset versions and check for duplicates, overlap between sources, text lengths, and label definitions. We will not treat manipulation labels as labels of truth or falsehood. We may also look for more datasets.

Data statement. The project will use public news datasets, and we do not plan to collect private data. We will record the licence terms for

[1] https://slavicnlp.github.io/disinformation-detection.html

Study	Data / task	Method and reported finding	Relevance / qualification
MIPD (2024)	Polish articles; binary disinformation	HerBERT and Polish RoBERTa show strong supervised results (Table 2).	Main Polish benchmark; random split.
PCoT (2025)	Five English test sets, including MultiDis	Prompts about persuasion improve the average F1 score (Table 2).	Direct baseline; tested only in English.
MALINT (2026)	English intent labels; multilingual disinformation tests	Reasoning about intent improves zero-shot detection. Encoders still work well for multilabel intent.	Binary intent, multilabel intent, and disinformation are separate tasks.
EU DisinfoTest (2024)	Credible and misleading narratives	Emotional language and appeals to authority affect zero-shot LLMs.	Shows that rhetorical framing can bias models.
F3 (2023)	Human and synthetic news and posts; in- and out-of-distribution	Zero-shot semantic reasoning transfers better than several specialised detectors.	Uses older models; useful for studying transfer direction.
Su et al. (2024)	Human- and machine-written news	Training on human-written data transfers to machine-written fake news better than the reverse.	Synthetic data must be tested carefully.
FADED (2025)	New-MultiFC and FND3; unseen domains	Adversarial training for documents and entities improves detection in unseen domains.	Entities may create shortcuts between text and labels.
TripleFact (2025)	Standard and contamination-controlled verification benchmarks	The framework separates memorisation from real detection.	Supports time-based tests and careful contamination checks.
SSA (2025)	LIAR2; controlled contamination	Changes to named entities reveal contamination linked to overly high accuracy.	An evaluation audit, not a new disinformation classifier.
MisBench (2025)	Conflicting knowledge and misinformation written in different styles	Reconstruct-to-Discriminate improves detection, but writing style is still a problem.	A broad test benchmark; not the same task as MIPD.
Wang et al. (2025)	LLM-generated fake news edited by people	Fake news written by people or LLMs remains hard to detect.	A controlled competition, not a shared-task ranking.
Irnawan et al. (2025)	Three non-English COVID-19 datasets	LLM agents use English evidence to make and explain predictions.	Evidence-based checking needs extra input and retrieval.
UNLP task (2025)	Manipulation in Ukrainian Telegram posts	Expert labels support classification of techniques and detection of text spans.	A related Slavic task; labels do not directly show truth or falsehood.
GerDISDETECT (2024)	German articles; semantic descriptions	An XLM-R model adapted to the domain provides multilabel baselines.	Uses a different label system from binary disinformation.
Calvo & Laureano (2026)	ISOT, Misinfo, and WELFake; transfer in both directions	Performance drops strongly when the data distribution changes. Transfer is uneven, while emotional features improve stability.	High accuracy on one dataset does not prove that a model will generalise.

Table 1: Fifteen published studies and their main results. The table clearly marks related tasks and settings that cannot be compared directly.

Model / source	Dataset and evaluated partition	Published metric and value
Polish RoBERTa-large (Modzelewski et al., 2024), Table 2	Official MIPD test set with 1,521 articles; mean from five training seeds	Weighted F1 0.960 ± 0.001; positive-class F1 0.930 ± 0.002
PCoT (Modzelewski et al., 2025), Table 5	Combined results from CoAID, ECTF, EUDisinfo, ISOT, and MultiDis; average for five LLMs and three prompt variants	F1: base 0.711; PCoT 0.815 (a difference of 0.104, not 15 percentage points)
RoBERTa-base / Llama 3.3 70B (Modzelewski et al., 2026), Table 4	MALINT test set with 500 articles; multilabel *intent* classification	Weighted F1: 0.821 / 0.570; these are not binary disinformation scores

Table 2: Published results. The rows cannot be ranked directly because they use different labels, metrics, and data splits.

Dataset / status	Size, class balance, and label definition	Acquisition / planned use	Reported licence
MIPD / READY	15,356 PL articles: 4,997 *disinformation* (32.5%, intentional manipulation/falsehood verified by fact-checkers) and 10,359 *credible* (67.5%, mainstream factual reporting).	Author repository; use official supervised split and test cross-lingual transfer to Polish.	CC BY-NC-ND 4.0
MultiDis / READY	1,922 EN articles: 630 *disinformation* (32.8%, operationally merges false/misleading claims and disinformation), 1,256 *credible* (65.3%), and 36 *ambiguous/off-topic* (1.9%).	PCoT repository; retain 1,886 binary-labelled records (excluding ambiguous) before deduplication.	CC BY-NC-ND 4.0 (paper)
MALINT / READY	1,600 EN articles: 584 *disinformation* (36.5%) and 1,016 *credible* (63.5%), plus 5 auxiliary multilabel malicious-intent categories (e.g., harm, polarisation).	Author repository; use official 770/330/500 train/dev/test split after checking overlap with MultiDis.	CC BY 4.0 (data); MIT (code)

Table 3: Planned datasets, class balances, label definitions, and readiness status as of 2 November 2026 (PL = Polish, EN = English).

code and data separately. News articles may name real people, so we will inspect the data before stating that it contains no personal data. We will keep only the fields we need and remove identifying details from examples. The data will stay outside the public code repository. We will provide download instructions and document IDs so that others can reproduce our work. The repository's code licence will not replace the dataset licences.

4 Solution and experimental plan

Figure 1 shows the planned pipeline. It includes a majority baseline, logistic regression with term frequency–inverse document frequency (TF-IDF), a fine-tuned XLM-R model, and one instruction-tuned LLM. We will test the LLM with plain prompts and prompts about persuasion or intent. After a small GPU-memory test, we will fix the model checkpoints and input limits before the final evaluation. Evidence retrieval is optional and will be tested separately if we use it.

Evaluation. The main metric will be F1. We will also report precision, recall, and confusion matrices. We will keep the official test IDs and report tests with held-out sources or topics separately. We will test transfer from English to Polish and between English datasets after removing duplicates. Source metadata and fact-check explanations will not be used as model inputs. If the organisers have not released their test data by Phase 2, we will use the existing datasets.

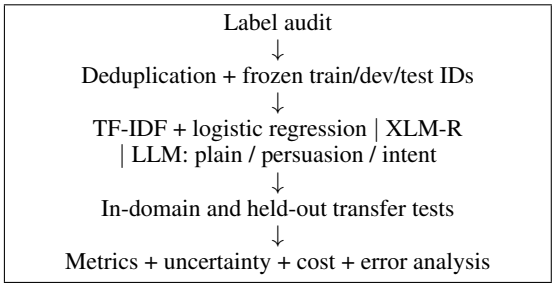


Figure 1: Proposed workflow. We will choose hyperparameters and prompts using only the development data.

We will run each experiment at least three times, each with a different random seed, and report the mean and standard deviation. We will also use paired bootstrap intervals when we compare test results. Invalid LLM answers will count as errors. We will document text truncation, the way chunks are combined, batch size, GPU type, peak memory, training and inference time, and token usage. We will inspect twenty errors and group them by rhetorical language, missing facts, domain shortcuts, and transfer problems.

Deliverables and schedule. The project has three main goals: (i) reproducible baselines from at least two model types, (ii) a comparison of results within one dataset and across datasets, and (iii) a direct comparison of prompt types that in-

cludes uncertainty and cost. A negative result will still be useful. We do not assume that the new methods will improve the scores. In the table, P means Piotr Bartosiewicz, I means Igor Lechoszest, and M means Mateusz Małachowski.

Date	Work and proposed owners
2–8 Nov	Literature and data deep dive / P: literature; I: data; M: summary
9–17 Nov	Mid-term preparation / P: slides; I: PoC (Proof of Concept); M: presentation
18 Nov	Mid-term presentation / all
19 Nov–1 Dec	Reproducible PoC and Phase 2 report / I: code; M: EDA (Exploratory Data Analysis) and results; P: report
2 Dec	Phase 2 report submission / all
3–23 Dec	Final experiment setup based on feedback / I: pipeline; M: evaluation; P: report plan
24 Dec–6 Jan	Winter break; no planned work / all
7–12 Jan	Experiments and final deliverables / I: runs; M: analysis; P: report
13 Jan	Project deadline / all

5 Limitations and declarations

The labels only approximate the author's intent, and removing unclear examples makes the results less general. A split by source or time cannot prove that a model did not see similar data during pretraining. Tests in a few languages cannot show that the method works for every language. If an API model is unavailable, we will use an open-weight LLM. We have not received reviewer feedback yet and will answer any later comments.

Name	**Contribution**	**Time (h)**
Piotr Bartosiewicz	Report creation and formatting	3
Igor Lechoszest	Report review and comments	3
Mateusz Małachowski	Articles and data research	3

Table 4: Contribution Record Table

LLM-use declaration. We used ChatGPT Codex to help find relevant papers. The team checked all claims and references before submission.

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